Do We Really Need Difficulty-Aware Sampling? The Label Distribution Confound in Knowledge Distillation

Anonymous ACL submission

Abstract

Difficulty-aware active learning is widely adopted in LLM-to-SLM distillation to maximize learning signal per annotation. We identify a confound that undermines this practice: for output-biased students, sample uncertainty and counter-prior-class membership are tightly coupled, so selecting uncertain samples silently distorts the training label distribution. We formalize this via the **Entanglement Score** $\varepsilon_{\text{ent}} = q_{\text{defer}} - q_{\text{accept}}$, a zero-cost diagnostic measuring the label-composition gap between uncertain and confident regions, and show that label distribution can dominate downstream performance over difficulty signal. **PCSS** separates distribution alignment from within-stratum uncertainty as ordered objectives, achieving the best Macro-F1 in 6 of 8 settings while never underperforming random sampling.

1 Introduction

Knowledge distillation from LLMs to SLMs (Hinton et al., 2015) is a standard deployment paradigm: a teacher annotates a curated subset and a student is fine-tuned on those labels. Under limited budgets, active learning prioritizes uncertain or informative samples (Settles, 2009), and recent distillation systems (Zhang et al., 2024; Liu et al., 2024) adopt this directly, expecting difficult samples to yield strictly better students.

However, this expectation fails for SLMs with severe output bias. Before fine-tuning, many SLMs predict a single label on nearly all inputs, concentrating uncertainty on counter-prior-class examples: a yes-biased model "hesitates" primarily on negative inputs. Uncertainty selection thus shifts training labels away from truth rather than targeting genuinely difficult regions. We term this the **difficulty–label confound**: a systematic coupling between what the student finds difficult and which class a sample belongs to (Figure 1).

We formalize the confound via the **Entanglement Score** ε_{ent}, a zero-cost diagnostic from existing teacher labels, and show that label distribution can dominate difficulty signal for biased SLMs. We propose **Prior-Corrective Stratified Selection (PCSS)**, decomposing selection into distribution alignment and within-stratum uncertainty as ordered objectives. Our contributions: **(1)** identifying the difficulty–label confound and introducing ε_{ent} as its diagnostic (§4); **(2)** proposing PCSS, best in 6/8 settings with asymmetric risk (§5–6); **(3)** showing CRC (Angelopoulos et al., 2024) serves dual roles as diagnostic lens and deployment certifier with distribution-free guarantees.

2 Related Work

Our work intersects active learning, knowledge distillation, and conformal prediction. Existing active learning methods prioritize model uncertainty (Lewis and Gale, 1994; Ash et al., 2020; Houlsby et al., 2011; Sener and Savarese, 2018), but none addresses the *label* distribution distortion induced by the student's output priors—distinct from class-imbalanced acquisition (Aggarwal et al., 2020), covariate shift (Farquhar et al., 2021), and curriculum ordering (Wang et al., 2022). Details in Appendix A.

3 Study Design

3.1 Task and Data Partitioning

We study active LLM-to-SLM distillation for binary classification. Given pool $\mathcal{U}_{\text{pool}}$, teacher $\mathcal{T}$, student θ_0, and budget B, the goal is to select $S \subset \mathcal{U}_{\text{pool}}$ ($|S|{=}B$), obtain teacher labels, and fine-tune θ_0 to produce θ^*. Two disjoint auxiliary sets support the framework: a teacher-labeled *guide set* $\mathcal{D}_{\text{guide}}$ drawn by stratified random sampling, and a held-out *certification set* $\mathcal{D}_{\text{cert}}$, both fixed before training to ensure $\mathcal{D}_{\text{cert}} \perp \theta^*$ (Appendix B).

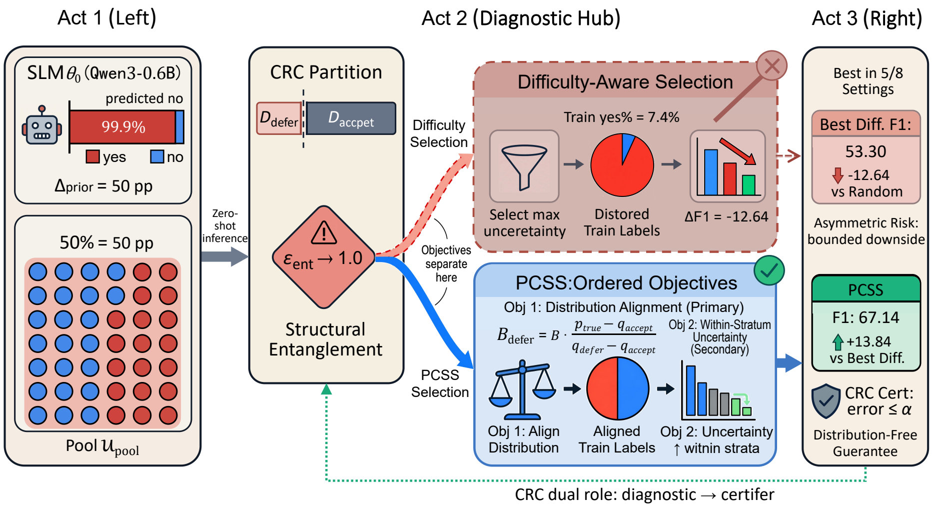

Figure 1: Framework overview. **Act 1**: A biased SLM (Δ_{prior}=50 pp) produces a CRC partition where uncertainty and label identity are structurally entangled (ε_{ent}→1.0). **Act 2**: Difficulty-aware selection draws from the uncertain region, distorting train labels (yes%=7.4%); PCSS first aligns the distribution (Obj 1), then selects uncertain samples within strata (Obj 2). **Act 3**: PCSS achieves best F1 in 6/8 settings with asymmetric risk; CRC provides a distribution-free deployment guarantee.

3.2 CRC-Guided Routing

We define a direction-agnostic certainty score from the student's logit margin $\ell_i = \log p_\theta(\text{“1”} \mid x_i) - \log p_\theta(\text{“0”} \mid x_i)$:

$$R_i(T) = \sigma(|\ell_i|/T) \in [0.5,\ 1], \tag{1}$$

where T is temperature and σ the logistic function. The absolute value ensures R_i reflects certainty regardless of predicted class.

We calibrate threshold $\hat{\lambda}$ on $\mathcal{D}_{\text{guide}}$ via Conformal Risk Control (CRC; Angelopoulos et al., 2024), providing finite-sample wrong-accept risk guarantees. With per-sample loss $L_j(\lambda) = \mathbf{1}\{R_j \geq \lambda\} \cdot \mathbf{1}\{\hat{y}_j \neq y_j\}$:

$$\hat{\lambda} = \min\left\{\lambda \in \Lambda : \tfrac{n_g}{n_g+1}\,\hat{R}_g(\lambda) + \tfrac{1}{n_g+1} \leq \alpha\right\}, \tag{2}$$

where $\frac{1}{n_g+1}$ is a finite-sample correction (Appendix E). This partitions the pool into uncertain $\mathcal{D}_{\text{def}}=\{x : R(x,T)<\hat{\lambda}\}$ and confident $\mathcal{D}_{\text{acc}}=\{x : R(x,T)\geq\hat{\lambda}\}$.

3.3 Difficulty-Aware Sampling Methods

We evaluate three difficulty-aware methods (full derivations in Appendix L): **CRC-error-mass** allocates budget between defer/accept sets via error concentration, with uniform random within-partition selection. **NS-error-mass** ranks all pool samples by embedding-weighted neighborhood accuracy, calibrated via isotonic regression (Zadrozny and Elkan, 2002), with importance-weighted PPS sampling. **Deferkcenter** uses CRC-error-mass allocation but replaces uniform selection within the defer set with k-Center Greedy (Sener and Savarese, 2018) for diversity.

3.4 Tasks and Models

We evaluate on seven binary tasks across four domains (Table 1): sentiment (IMDb (Maas et al., 2011), 2 queries), hate speech (TwitterHate (Basile et al., 2019)), fact verification (FEVER (Thorne et al., 2018)), and programmer profiling (Codebase, 3 queries). The primary student is Qwen3-0.6B; we additionally evaluate Qwen3-1.7B on FEVER as a capacity contrast (Yang et al., 2025). The teacher is GPT-4o-mini (OpenAI, 2024). Hyperparameters are in Appendix B.

4 Diagnostic Analysis

Output prior bias is pervasive. We define $\Delta_{\text{prior}} = |\Pr[\hat{y}{=}1] - \hat{p}_1|$. Table 1 shows Qwen3-0.6B achieves Macro-F1 <46 on all 7 tasks with $\Delta_{\text{prior}} \in [16.8, 87.6]$ pp. The 1.7B model on FEVER (Δ_{prior}=6.5 pp, F1= 86.72) serves as a low-bias contrast.

Task	N	True %	Pred %	F1	Δ_{prior} (pp)
IMDb q1	50K	50.0	99.99	33.35	50.0
IMDb q2	50K	27.5	100.0	21.57	72.5
TwitterHate	17K	83.2	100.0	45.42	16.8
Codebase q1	9.3K	6.3	93.82	12.30	87.6
Codebase q2	9.3K	62.0	99.45	39.60	37.5
Codebase q3	9.3K	25.8	99.88	20.65	74.1
FEVER (0.6B)	165K	52.4	10.01	44.08	42.4
FEVER (1.7B)	*165K*	*52.4*	*58.9*	*86.72*	*6.5*

Table 1: Zero-shot output prior bias. Pred % is fraction predicted positive. *Italic*: capacity-contrast (1.7B).

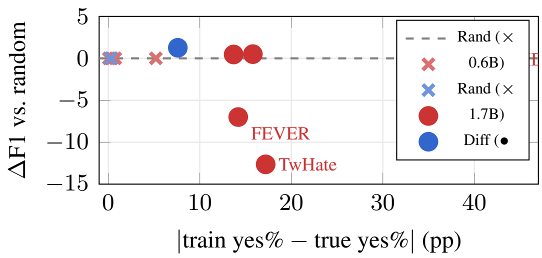

Figure 2: Training label gap vs. ΔF1. Random baselines (×) cluster near origin; difficulty methods (●) with larger gaps show degraded performance.

The difficulty–label confound. When a model assigns near-uniform $\hat{y}{=}1$, only counter-prior inputs produce low R_i, landing in $\mathcal{D}_{\text{def}}$. The CRC partition thus simultaneously functions as a *label* partition: uncertain samples are systematically enriched in one class.

To quantify this, let q_{defer} and q_{accept} denote positive-label fractions in $\mathcal{D}_{\text{guide}} \cap \mathcal{D}_{\text{def}}$ and $\mathcal{D}_{\text{guide}} \cap \mathcal{D}_{\text{acc}}$, computable at zero additional cost. The **Entanglement Score**:

$$\varepsilon_{\text{ent}} = q_{\text{defer}} - q_{\text{accept}}, \tag{3}$$

measures the label-composition gap between uncertain and confident regions. Under uniform within-stratum sampling with defer fraction s_{def}:

$$\mathbb{E}[p_S^{(1)}] - \hat{p}_{\text{pool}} = (s_{\text{def}} - r_U) \cdot \varepsilon_{\text{ent}}, \tag{4}$$

where $r_U = |\mathcal{D}_{\text{def}}|/|\mathcal{U}_{\text{pool}}|$ (derivation in Appendix F). Under non-uniform selection (e.g., PCSS), the additive residual is $\leq$0.03 empirically (Appendix H).

Label distribution can dominate performance. Table 2 shows all three difficulty methods shift training labels by 14–43 pp in their worst cases. Figure 2 confirms a negative trend between label gap and ΔF1. A controlled experiment on Codebase q1 (Table 3) fixes the sample source and varies only label composition: the 15.44 F1 gap demonstrates that distribution shifts of the magnitude induced by difficulty methods (14–43 pp) far exceed difficulty-driven gains ($\leq$1.26 F1). The monotonic trend in Figure 2 suggests the effect persists at moderate shifts.

Capacity moderation. On FEVER, the 0.6B model (Δ_{prior}=42.4 pp) gains at most +0.50 F1 from difficulty with 15.8 pp label shift; the 1.7B model (Δ_{prior}=6.5 pp, $|\varepsilon_{\text{ent}}|\approx$0.05) gains +1.26 with only 7.6 pp shift. As $|\varepsilon_{\text{ent}}| \to 0$, the confound attenuates and PCSS converges to proportional sampling.

Task	Method	Train yes%	True yes%	Eval pred%	F1
TwHate	random	78.0	83.2	87.4	65.94
	crc-error-mass	66.0	83.2	45.4	53.30
IMDb	random	50.8	50.0	50.5	94.37
	ns-error-mass	7.4	50.0	45.6	92.25
FEVER†	random	52.0	52.4	71.8	76.47
	defer-kcenter	66.6	52.4	80.0	69.45

Table 2: Worst-case failures for each difficulty method. †: Pos-F1 (Appendix D).

Implication. Difficulty-aware selection conflates *distributional fidelity* and *informative sampling*. For biased SLMs these are anti-correlated: pursuing informativeness distorts the distribution, and distributional fidelity dominates performance. This motivates separating and ordering the two objectives.

5 Prior-Corrective Stratified Selection

5.1 Ordered Objectives

PCSS decomposes selection into two strictly ordered objectives.

Obj 1 (primary): Distribution alignment. $\min_S |p_S^{(1)} - \hat{p}_{\text{true}}|$, where $\hat{p}_{\text{true}} = \frac{1}{n_g}\sum_{j\in\mathcal{D}_{\text{guide}}} y_j^{\mathcal{T}}$ is the true positive rate estimated from teacher labels on $\mathcal{D}_{\text{guide}}$.

Obj 2 (secondary): Within-stratum uncertainty. Conditional on Obj 1, maximize $\sum_{i\in S_c}(1{-}R_i(T))$ within each stratum c.

5.2 Algorithm

Guide-based estimation. PCSS estimates q_{defer} and q_{accept} from teacher labels on $\mathcal{D}_{\text{guide}}$ at zero additional cost, avoiding reliance on the student's vacuous proxy labels. $\mathcal{D}_{\text{guide}}$ serves multiple roles (CRC calibration, ε_{ent} computation, $\hat{p}_{\text{true}}$ estimation); potential circularity is discussed in Appendix J.

Condition	Train yes%	F1	Δ
artificially-balanced	50.0	**57.93**	—
natural-distribution	0.8	42.49	−15.44
wrong-source-natural	100.0	26.77	−31.16

Table 3: Distribution sensitivity on Codebase q1 (true rate 6.3%). Same source, different label composition (rows 1 vs. 2): 15.44 F1 gap. Neither condition matches the true rate; the experiment measures sensitivity, not optimality.

Task	n	Random	Best Diff.	PCSS
TwitterHate	50	65.94	53.30 (−12.64)	**67.14** (+1.20)
	125	55.70	65.85 (+10.15)	**68.62** (+12.92)
	250	57.62	**71.16** (+13.54)	67.83 (+10.21)
	2231	90.42	90.60 (+0.18)	**91.08** (+0.66)
FEVER 0.6B	1500	90.07	90.52 (+0.45)	**90.58** (+0.51)
	3000	93.85	**94.20** (+0.35)	94.03 (+0.18)
IMDb q1	2500	94.37	92.25 (−2.12)	**94.83** (+0.46)
Codebase q2	500	66.81	58.23 (−8.58)	**69.37** (+2.56)

Table 4: Macro-F1 (Δ vs. random). Best Diff. is oracle max across three difficulty methods. PCSS is best in 6/8 settings.

Budget allocation. From Eq. 4, targeting $\mathbb{E}[p_S^{(1)}] = \hat{p}_{\text{true}}$ yields (Appendix G):

$$B_{\text{def}} = \text{clip}\left(\text{round}\left(B \cdot \frac{\hat{p}_{\text{true}} - q_{\text{accept}}}{\varepsilon_{\text{ent}}}\right),\ 0,\ B\right). \quad (5)$$

Finite-sample consistency is formalized in Proposition **??**. When $|\varepsilon_{\text{ent}}| < \delta$, PCSS defaults to proportional allocation $B_{\text{def}} = \text{round}(B \cdot r_U)$; no fallback was triggered in any of the 8 main settings (Appendix I).

Within-stratum selection. Within each partition, the B_c lowest-$R(x, T)$ samples are selected (Obj 2). Replacing Obj 2 with uniform sampling recovers Balanced-Random (Obj 1 only); setting B_{def}=B recovers difficulty-only.

5.3 Deployment Certification

CRC serves dual roles: diagnostic (threshold $\hat{\lambda}$ on $\mathcal{D}_{\text{guide}}$) and deployment certifier (threshold λ_c^* on $\mathcal{D}_{\text{cert}}$). Since $\mathcal{D}_{\text{cert}} \perp \theta^*$:

$$\mathbb{E}[\mathbf{1}\{R(Z_{\text{new}}) \geq \lambda_c^*\} \cdot \mathbf{1}\{\hat{y}(Z_{\text{new}}) \neq y(Z_{\text{new}})\}] \leq \alpha. \quad (6)$$

This holds for any selection strategy used to construct θ^* (Appendix **??**).

6 Experiments

Setup. We evaluate PCSS on 4 tasks across 8 budget settings (Table 4). "Best Diff." is the oracle maximum across all three difficulty methods (per-method breakdowns in Appendix **??**). All values are mean Macro-F1 over 3 seeds (SD ≤2.1; Appendix N).

Results. Where difficulty *underperforms* random (TwHate n=50, IMDb, Codebase), PCSS outperforms both, recovering up to +13.84 F1 over the best difficulty method. Across all 8 settings, PCSS ≥ Random (min Δ: +0.18), confirming distribution alignment provides a performance floor.

The risk profile is asymmetric: where difficulty wins (2 settings), its margin over PCSS is ≤3.33; where PCSS wins, the margin reaches 13.84.

Medium-budget regime. At TwitterHate n=250, difficulty surpasses PCSS by 3.33 F1. As budget grows, expected label shift shrinks as $O(1/\sqrt{n})$, reducing Obj 1's marginal value while Obj 2 contributes genuine information gain. Adaptive objective weighting is a promising extension.

Capacity comparison. On FEVER, the 0.6B model gains +0.50 F1 with 15.8 pp label shift, while the 1.7B gains +1.26 with only 7.6 pp shift, consistent with the confound diminishing as $|\varepsilon_{\text{ent}}|$ decreases.

7 Conclusion

We identify a confound in difficulty-aware selection for LLM-to-SLM distillation: for biased students, uncertainty and counter-prior-class membership are tightly coupled, causing difficulty selection to distort training label distributions. The Entanglement Score $\varepsilon_{\text{ent}} = q_{\text{defer}} - q_{\text{accept}}$ directly governs how allocation decisions translate to distribution deviation (Eq. 4). PCSS resolves the confound by ordering distribution alignment above uncertainty selection, achieving the best Macro-F1 in 6/8 settings with asymmetric risk (max downside 3.33, max upside 13.84).

Limitations

Task scope. Our study is restricted to binary classification; extending ε_{ent} to multi-class settings requires measuring per-class label-composition gaps, and whether per-class effects compound or cancel is open.

Model diversity. We evaluate only Qwen3 (0.6B primary, 1.7B on FEVER). Verifying across architectures (LLaMA, Phi, Gemma) and confirming graceful degradation when $|\varepsilon_{\text{ent}}| \approx 0$ are important next steps.

Baselines. We compare against three uncertainty-based methods. BADGE (Ash et al., 2020) and BALD (Houlsby et al., 2011) share the uncertainty-selection principle but differ in operationalization; confirming the confound persists under these mechanisms would strengthen generality.

Statistical power. Results averaged over 3 seeds; effects below $\sim$2 F1 (e.g., FEVER +0.51) fall within seed variance and should be interpreted cautiously.

Diagnostic robustness. ε_{ent} depends on T and α; Appendix M shows weak sensitivity on one task; multi-task verification is needed.

Causal evidence. The isolation experiment (Table 3) uses one task at an extreme shift; replication across tasks and intermediate magnitudes would improve generalizability.

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A Related Work

Active learning. Pool-based active learning selects samples to maximize learning efficiency under a labeling budget. Uncertainty sampling (Lewis and Gale, 1994) selects samples with highest predictive entropy or smallest margin. BADGE (Ash et al., 2020) combines gradient embeddings with k-means++ for diversity. BALD (Houlsby et al., 2011) maximizes mutual information between model parameters and predictions. CoreSet (Sener and Savarese, 2018) formulates selection as a geometric covering problem. All share the premise that model uncertainty indicates learning value. Our work shows this premise can be violated when the student exhibits output prior bias, a failure mode orthogonal to the algorithmic differences among these methods.

Knowledge distillation with active selection. Recent work applies active learning to LLM-to-SLM distillation (Zhang et al., 2024; Liu et al., 2024), selecting difficult or uncertain samples for teacher annotation. These systems inherit the uncertainty-selection premise without accounting for the student's output prior, making them susceptible to the confound we identify.

Class imbalance and distribution shift. Class-imbalanced active learning (Aggarwal et al., 2020) addresses settings where the pool itself is skewed; our finding differs because even a perfectly balanced pool (e.g., IMDb at 50/50) suffers label distortion when the student is biased. Pool-based distribution shift (Farquhar et al., 2021) concerns covariate shift between pool and test distributions; our confound is a *label* distribution distortion *within* the selected training set, induced by the student's own output priors.

Conformal prediction. CRC (Angelopoulos et al., 2024) provides distribution-free risk guarantees for selective prediction. We use CRC in a dual capacity: as a diagnostic tool (threshold $\hat{\lambda}$ on $\mathcal{D}_{\text{guide}}$) for the defer/accept partition, and as a deployment certifier (threshold λ_c^* on $\mathcal{D}_{\text{cert}}$).

B Experimental Setup

Data splits. From each task, n_g=1000 samples are drawn for $\mathcal{D}_{\text{guide}}$ and n_c=200 for $\mathcal{D}_{\text{cert}}$ via stratified random sampling (seed-fixed), both teacher-labeled before any model training. The remainder forms $\mathcal{U}_{\text{pool}}$. Stratified sampling ensures $\hat{p}_{\text{true}}$ is an approximately unbiased estimator of $\hat{p}_{\text{pool}}$.

CRC hyperparameters. Risk level α=0.1; temperature T=1 (sensitivity in Appendix M). Threshold grid: $\Lambda = \{0.50, 0.51, \ldots, 1.00\}$. If no λ satisfies Eq. 2, we set $\hat{\lambda}$=1.01 (all defer).

PCSS hyperparameters. $\delta = 0.02$ for the $|\varepsilon_{\text{ent}}|$ fallback threshold; minimum overlap $|\mathcal{D}_{\text{guide}} \cap \mathcal{D}_{\text{def}}| \geq 30$ for reliable q_{defer} estimation. See Appendix I for justification.

Training. LoRA (Hu et al., 2022) (rank 1, α_{LoRA}=16, dropout 0.05) on all attention and MLP projections. Learning rate 10^{-4}, 3 epochs, effective batch size 16, cosine schedule, bf16 precision. Each round trains from the base model (not continual) to preserve $\mathcal{D}_{\text{guide}} \perp \theta^*$.

Evaluation. Macro-F1 (mean of per-class F1) unless otherwise noted. All results are averaged over 3 random seeds.

C Task Details

IMDb (Maas et al., 2011): 50K movie reviews; q1 tests positive sentiment, q2 tests discussion of acting quality. **TwitterHate** (Basile et al., 2019): 17K tweets annotated for hate speech. **FEVER** (Thorne et al., 2018): 165K claim–evidence pairs for fact verification. **Codebase**: 9.3K developer–repository pairs; q1 tests contribution history, q2 tests project interest, q3 tests code review activity.

D Metric Details

For FEVER in Table 2, Positive-F1 (F1 of the positive class) is reported because the evaluation subset targets a specific class of interest.

E CRC Calibration: Theory

Routing score. $R_i(T) = \sigma(|\ell_i|/T) = 1/(1 + \exp(-|\ell_i|/T)) \in [0.5, 1]$. The threshold $\hat{\lambda}$ corresponds to a margin cutoff:

$$R_i(T) \geq \hat{\lambda} \iff |\ell_i| \geq T \cdot \log \frac{\hat{\lambda}}{1-\hat{\lambda}} \triangleq \tau_{\text{crc}}. \quad (7)$$

CRC loss. $L_j(\lambda) = \mathbf{1}\{R_j \geq \lambda\} \cdot \mathbf{1}\{\hat{y}_j \neq y_j\}$. Population risk: $R(\lambda) = \mathbb{E}[L(Z;\lambda)]$; empirical risk: $\hat{R}_g(\lambda) = \frac{1}{n_g}\sum_{j=1}^{n_g} L_j(\lambda)$.

Finite-sample guarantee. Under the learn-then-test framework (Angelopoulos et al., 2025), the corrected bound $\hat{R}_g^+(\lambda) = \frac{n_g}{n_g+1}\hat{R}_g(\lambda) + \frac{1}{n_g+1}$ satisfies $\mathbb{E}[L(Z_{\text{new}};\hat{\lambda})] \leq \alpha$ whenever $\hat{\lambda} = \min\{\lambda \in \Lambda : \hat{R}_g^+(\lambda) \leq \alpha\}$. Since $L_j(\lambda)$ is non-increasing in λ, ascending search finds the smallest valid threshold.

F Entanglement Score: Derivation

Let $r_U = |\mathcal{D}_{\text{def}}|/|\mathcal{U}_{\text{pool}}|$. The pool positive rate decomposes as:

$$\hat{p}_{\text{pool}} = r_U \cdot q_{\text{defer}} + (1-r_U) \cdot q_{\text{accept}}. \quad (8)$$

Under stratified sampling with defer fraction $s_{\text{def}} = B_{\text{def}}/B$ and uniform within-stratum selection:

$$\mathbb{E}[p_S^{(1)}] = s_{\text{def}} \cdot q_{\text{defer}} + (1-s_{\text{def}}) \cdot q_{\text{accept}}. \quad (9)$$

Subtracting:

$$\mathbb{E}[p_S^{(1)}] - \hat{p}_{\text{pool}} = (s_{\text{def}} - r_U) \cdot \varepsilon_{\text{ent}}. \quad (10)$$

Per-task values are in Table 5.

Task	q_{defer}	q_{accept}	ε_{ent}
IMDb q1	0.03	0.52	−0.49
IMDb q2	0.01	0.29	−0.28
TwitterHate	0.42	0.91	−0.49
Codebase q1	0.02	0.07	−0.05
Codebase q2	0.18	0.65	−0.47
Codebase q3	0.03	0.27	−0.24
FEVER 0.6B	0.89	0.47	+0.42
FEVER 1.7B	*0.48*	*0.53*	*−0.05*

Table 5: Per-task entanglement scores.

G PCSS Budget Derivation

PCSS targets $\mathbb{E}[p_S^{(1)}] = \hat{p}_{\text{true}}$. Since $\mathcal{D}_{\text{guide}}$ is drawn from $\mathcal{U}_{\text{pool}}$ by stratified sampling, $\hat{p}_{\text{true}} \approx \hat{p}_{\text{pool}}$. Substituting into Eq. 9 and solving:

$$s_{\text{def}} = \frac{\hat{p}_{\text{true}} - q_{\text{accept}}}{\varepsilon_{\text{ent}}}. \quad (11)$$

The budget is $B_{\text{def}} = \text{round}(B \cdot s_{\text{def}})$, clipped to $[0, B]$.

Finite-sample error propagation. PCSS uses estimates $\hat{q}_{\text{def}}, \hat{q}_{\text{acc}}$ in place of $q_{\text{defer}}, q_{\text{accept}}$, yielding $\hat{s}_{\text{def}} = (\hat{p}_{\text{true}} - \hat{q}_{\text{acc}})/\hat{\varepsilon_{\text{ent}}}$. The allocation error:

$$\hat{s}_{\text{def}} - s^*_{\text{def}} = \frac{-\Delta q_{\text{acc}} - s^*_{\text{def}} \cdot \Delta\varepsilon_{\text{ent}}}{\hat{\varepsilon_{\text{ent}}}}, \quad (12)$$

where $\Delta q_{\text{acc}} = \hat{q}_{\text{acc}} - q_{\text{acc}}$, $\Delta\varepsilon_{\text{ent}} = \Delta q_{\text{def}} - \Delta q_{\text{acc}}$. Under $|\hat{\varepsilon_{\text{ent}}}| \geq \delta$ and Hoeffding bounds (Hoeffding, 1963) $|\Delta q_c| = O(n_c^{-1/2})$: $|\hat{s}_{\text{def}} - s^*_{\text{def}}| = O(n_{\min}^{-1/2}/\delta)$.

H Within-Stratum Selection Bias

Eq. 10 holds exactly under uniform within-stratum sampling. Under PCSS's Obj 2 (selecting lowest-R samples), the selected positive rates $q^{\text{sel}}_{\text{defer}}$, $q^{\text{sel}}_{\text{accept}}$ may differ from the stratum averages. The additive residual:

$$\text{bias}_{\text{res}} = s_{\text{def}}(q^{\text{sel}}_{\text{defer}} - q_{\text{defer}}) + (1-s_{\text{def}})(q^{\text{sel}}_{\text{accept}} - q_{\text{accept}}). \quad (13)$$

Empirically, $|\text{bias}_{\text{res}}| \leq 0.03$ across all settings—an order of magnitude smaller than the cross-stratum deviation prevented by Eq. 5 (typically 10–40 pp). A formal bound would require a regularity condition (e.g., Lipschitz continuity of the label rate as a function of R within each stratum) that we do not impose.

I PCSS Edge Cases

Threshold choices. $\delta = 0.02$ for $|\varepsilon_{\text{ent}}|$; sample-size threshold of 30 for $|\mathcal{D}_{\text{guide}} \cap \mathcal{D}_{\text{def}}|$ (95% CI

$\leq \pm 0.18$ by Hoeffding). The choice $\delta = 0.02$ ensures that Proposition **??**'s error amplification $1/\delta = 50$ is manageable: with $n_{\min} \geq 30$, the allocation error is bounded by $\sim 50 \times 0.18 = 9$ pp, comparable to the shifts prevented. Smaller δ values would improve distribution alignment for weakly entangled tasks but amplify finite-sample noise.

Trigger frequency. In all 8 main settings and 3 appendix settings, no fallback was triggered ($|\varepsilon_{\text{ent}}| \geq 0.05$, defer overlap ≥ 47). The $[0, B]$ clip was never activated.

J Guide Set Multiple-Use Discussion

$\mathcal{D}_{\text{guide}}$ serves three roles: CRC calibration (determining $\hat{\lambda}$), ε_{ent} computation, and $\hat{p}_{\text{true}}$ estimation. The circularity is benign: the CRC partition depends on the student's *predictions*, while q_{defer}, q_{accept}, $\hat{p}_{\text{true}}$ depend on the teacher's *labels*—independent conditional on inputs $\{x_j\}$. The certification guarantee (Proposition **??**) is unaffected because it uses $\mathcal{D}_{\text{cert}}$. Sensitivity to n_g is a direction for future work.

K Causal Isolation: Discussion

Table 3 uses Codebase q1 (true rate 6.3%). The 49.2 pp shift between conditions is more extreme than the 14–43 pp induced by difficulty methods (Table 2). The experiment establishes that label distribution *can* dominate at these magnitudes but does not claim it always does. The monotonic trend in Figure 2 provides complementary evidence at moderate shifts.

L Method Derivations

L.1 CRC-Error-Mass

Diagnostic quantities. $r_U = |\mathcal{D}_{\text{def}}|/|\mathcal{U}_{\text{pool}}|$ (pool defer fraction), $r_C = \frac{1}{n_g}\sum_j \mathbf{1}\{R_j < \hat{\lambda}\}$ (guide defer fraction). Error rates on $\mathcal{D}_{\text{guide}}$:

$$e_{\text{all}} = \tfrac{1}{n_g}\textstyle\sum_{j=1}^{n_g} \mathbf{1}\{\hat{y}_j \neq y_j\}, \tag{14}$$

$$e_{\text{def}} = \frac{\sum_{j:\, R_j < \hat{\lambda}} \mathbf{1}\{\hat{y}_j \neq y_j\}}{\sum_j \mathbf{1}\{R_j < \hat{\lambda}\}}. \tag{15}$$

Boost and defer share. $c_{\text{crc}} = e_{\text{def}}/e_{\text{all}}$ (error concentration).

$$\eta = \text{clip}\left(\tfrac{\log c_{\text{crc}}}{\log(1/r_C)},\ 0,\ 1\right), \quad s_{\text{def}} = r_U + \eta(1 - r_U)^2. \tag{16}$$

The denominator $\log(1/r_C)$ tempers the boost when defer is rare. Budget: $B_{\text{def}} = \text{round}(B \cdot s_{\text{def}})$, $B_{\text{acc}} = B - B_{\text{def}}$, with uniform random within-partition selection.

L.2 NS-Error-Mass

Same-prediction neighbors $\mathcal{N}_i = \{j \in \mathcal{D}_{\text{guide}} : \hat{y}_j = \hat{y}_i\}$ with $w_{ij} = \max(\cos(z_i, z_j), 0)$. Local accuracy:

$$a_i = \frac{\sum_{j \in \mathcal{N}_i} w_{ij}\, \mathbf{1}\{\hat{y}_j = y_j\}}{\sum_{j \in \mathcal{N}_i} w_{ij}}. \tag{17}$$

Difficulty $d_i = 1 - a_i$ is calibrated to error probabilities p_i via isotonic regression (Zadrozny and Elkan, 2002). Accept error rate:

$$e_{\text{acc}} = \text{clip}\left(\frac{e_{\text{all}} - r_C \cdot e_{\text{def}}}{1 - r_C},\ 0,\ 1\right). \tag{18}$$

Target density $e_{\text{target}} = (1 - s_{\text{def}}) \cdot e_{\text{acc}} + s_{\text{def}} \cdot e_{\text{def}}$ (from Eq. 16). Weights $\tilde{w}_i(\beta) = ((p_i + \epsilon)/(e_{\text{all}} + \epsilon))^\beta$ with $\epsilon = 1/(n_g + 1)$, tuned via binary search over $\beta \in [0, 5]$, used for PPS sampling.

Under severe bias (>99% predicting one class), $\mathcal{N}_i \approx \mathcal{D}_{\text{guide}}$, rendering local accuracy indiscriminate and compounding the confound.

L.3 Defer-k-Center

Budget from CRC-error-mass. Within defer: initialize farthest from centroid; iteratively add $\arg\max_i \min_{j \in \text{sel}} \|z_i - z_j\|_2$.

M Hyperparameter Sensitivity

Varying $\alpha \in \{0.05, 0.10, 0.15\}$ on TwitterHate (n=125): $\leq$1.2 F1 variation. Varying $T \in \{0.5, 1.0, 2.0\}$: $\leq$0.8 F1 variation; $|\varepsilon_{\text{ent}}|$ varies by $\leq$0.06. Both are small relative to distribution effects (>10 F1). Multi-task sensitivity analysis would strengthen these findings.

N Per-Seed Variance

SD across 3 seeds: $\leq$2.1 F1 in all settings. Effects >10 F1 exceed 6× seed variance. Effects <2 F1 should be interpreted with caution.

O Excluded Tasks

Three tasks with $\hat{p}_1 < 26\%$ are excluded from main evaluation. PCSS $\geq$ Random in all cases.